- Stay alive, be active, move and work
- Build new cells and tissues for growth;
- Stay healthy and heal themselves;
- Prevent and fight infections.

Foods are classified according to their functions in the body. The functions of food can be broadly classified into three main categories

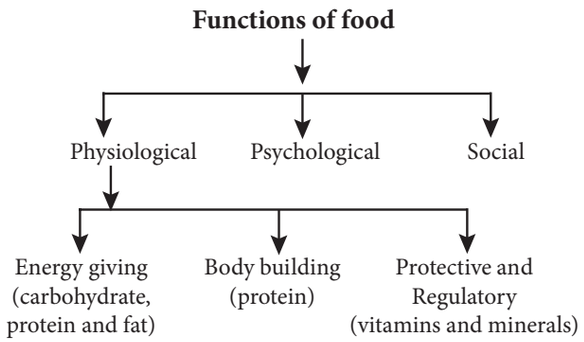


Fig 1.1: Functions of food

1.2. 1 Physiological functions of food

The physiological functions of food can be further sub-divided as follows:

a. Energy giving

b. Body building

c. Protective and Regulatory

a) Energy giving

This group includes foods rich in carbohydrate, fats and proteins. Energy is defined in terms of kilo calories and thus one gram of carbohydrate gives 4 kcal, one gram of protein gives 4 kcal, while one gram of fat gives 9 kcal. This group may be broadly divided into two groups:

- Cereals, pulses, nuts and oilseeds, roots and tubers.
- Simple carbohydrates like sugars, fats and oils.

In addition to energy cereals provide large amounts of protein, minerals and vitamins in the diet. Pulses also give protein and B vitamins besides giving energy to the body. Nuts and oilseeds are rich in energy yielding as they are good sources of fats and proteins. Roots and tubers though mainly provides energy, contribute to some extent to minerals and vitamins.

Simple carbohydrates like sugars provide only energy. Fats and oils provides fat,energy and fat soluble vitamins.

What are empty calories?

Simple carbohydrates like sugars provide only energy (empty calories) and do not provide any other nutrients.

b) Body Building:

The food we eat is a part of us. Thus one of the most important functions of food is that of building the body. They are classified into two groups:

- **Milk, egg, meat and fish:** They are rich in protein of high biological value. These proteins have all essential amino acids in correct proportion for the synthesis of body tissues.
- **Pulses, oilseeds and nuts:** They are rich in protein but may not contain all the essential amino acids required by the human body.

c) Protective and Regulatory function

Foods rich in protein, vitamins and minerals have regulatory functions in the

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